

1. Generative AI Fundamentals

Q1. What is Generative AI?

Answer:

Generative AI refers to AI systems capable of generating new content such as text, images, audio, video, and code based on patterns learned from training data.

Examples

- ChatGPT → Text generation
 - GitHub Copilot → Code generation
 - DALL-E → Image generation
 - Sora → Video generation
-

Q2. What is an LLM?

Answer:

Large Language Models (LLMs) are deep learning models trained on massive text datasets to understand and generate human-like language.

Examples

- OpenAI GPT-4/5
 - Anthropic Claude
 - Google Gemini
 - Llama
-

Q3. How does an LLM work?

Answer:

Flow:

```
Input Prompt
  ↓
Tokenization
  ↓
Transformer Architecture
  ↓
Attention Mechanism
  ↓
Probability Prediction
```

↓
Generated Output

Q4. What is Tokenization?

Answer:

Process of breaking text into smaller units called tokens.

Example:

"I love AI"

Tokens:
["I", "love", "AI"]

Q5. What is Context Window?

Answer:

Amount of information an LLM can remember during a conversation.

Example:

8K Tokens
32K Tokens
128K Tokens
1M Tokens

Q6. What are Hallucinations?

Answer:

When an LLM generates incorrect or fabricated information confidently.

Causes

- Missing knowledge
 - Ambiguous prompts
 - Lack of external grounding
-

Q7. What is Temperature?

Answer:

Controls randomness.

Temperature	Behavior
0	Deterministic
0.3	Stable
0.7	Creative
1+	Highly random

2. Prompt Engineering

Q8. What is Prompt Engineering?

Answer:

Designing prompts to obtain desired outputs from an LLM.

Q9. What are different prompting techniques?

Zero Shot

`Translate English to French.`

One Shot

`English: Hello
French: Bonjour`

`English: Good Morning`

Few Shot

Multiple examples provided.

Chain of Thought

`Think step by step.`

Q10. What is System Prompt?

Answer:

Instructions controlling model behavior.

Example:

You are a Data Engineering Expert.

3. Embeddings & Vector Databases

Q11. What are Embeddings?

Answer:

Numerical vector representation of text that captures semantic meaning.

Example:

Dog → [0.23, 0.44, 0.78...]

Puppy → [0.25, 0.41, 0.80...]

Semantically similar words have similar vectors.

Q12. Why are embeddings important?

Answer

Used for:

- Semantic Search
 - RAG
 - Recommendation Systems
 - Similarity Search
-

Q13. What is a Vector Database?

Answer

Database optimized for storing and searching embeddings.

Examples

- Pinecone
- Weaviate
- Milvus
- Chroma

- Qdrant

Q14. What is Similarity Search?

Answer

Finding nearest vectors using:

- Cosine Similarity
- Euclidean Distance
- Dot Product

4. RAG (Retrieval Augmented Generation)

Q15. What is RAG?

Answer

RAG retrieves relevant documents and injects them into the LLM context before generating a response.

Q16. Why RAG?

Answer

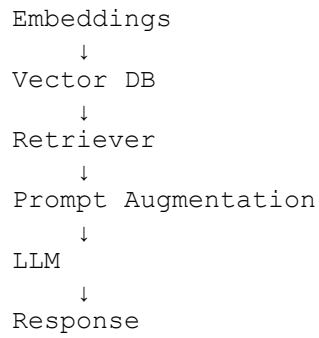
Solves:

- Hallucinations
- Knowledge Cutoff
- Enterprise Knowledge Access
- Private Data Access

Q17. Explain RAG Architecture.

Answer

Documents
↓
Chunking
↓



Q18. What is Chunking?

Answer

Breaking documents into smaller sections.

Types

- Fixed Size
- Recursive
- Semantic
- Sliding Window

Q19. What is Hybrid Search?

Answer

Combination of:

- Vector Search
- Keyword Search

Q20. What are RAG Evaluation Metrics?

Answer

- Context Precision
 - Context Recall
 - Faithfulness
 - Relevance
 - Groundedness
-

5. AI Agents

Q21. What is an AI Agent?

Answer

An AI system capable of:

- Reasoning
 - Planning
 - Tool Usage
 - Memory Management
 - Task Execution
-

Q22. Agent vs RAG?

RAG	Agent
Retrieves knowledge	Takes actions
Passive	Active
Search-oriented	Task-oriented
Single-step	Multi-step

RAG and agents are complementary, not competing technologies.

Q23. Agent Workflow?

Answer

```
Goal
↓
Reason
↓
Plan
↓
Use Tools
↓
Observe
↓
Repeat
↓
Final Answer
```

Q24. What is ReAct?

Answer

Reason + Act pattern.

Thought
Action
Observation
Thought
Action
Final Answer

6. Agentic AI

Q25. What is Agentic AI?

Answer

Systems where AI autonomously decides:

- What to do
- Which tool to use
- When to stop

without hardcoded workflows.

Q26. Characteristics of Agentic Systems

Planning

Memory

Tool Usage

Reflection

Self-Correction

Multi-Step Reasoning

Q27. What is Agent Memory?

Short-Term Memory

Conversation Context

Long-Term Memory

Vector Store / Database

7. LangChain

Q28. What is LangChain?

Answer

Framework for building LLM applications with:

- Prompts
 - Chains
 - Agents
 - Memory
 - Tools
 - RAG Integrations
-

Q29. Main Components of LangChain

- Prompt Templates
 - Models
 - Chains
 - Agents
 - Tools
 - Memory
 - Retrievers
-

Q30. When should LangChain be used?

Answer

Good for:

- Chatbots
- RAG
- Tool Calling
- Rapid Prototyping

8. LangGraph

Q31. What is LangGraph?

Answer

A stateful graph-based framework for agent orchestration that supports loops, branching, retries, checkpoints, and human-in-the-loop workflows.

Q32. Why LangGraph?

Answer

Traditional chains are linear.

LangGraph supports:

- Cyclic workflows
 - State management
 - Multi-agent systems
 - Durable execution
-

Q33. Core Concepts of LangGraph

Nodes

Tasks

Edges

Transitions

State

Shared memory

Conditional Routing

Decision making

Q34. LangChain vs LangGraph?

LangChain	LangGraph
Simple	Advanced
Linear	Graph
Stateless	Stateful
Easy	Production-grade

LangGraph is often used underneath advanced LangChain agents.

9. MCP (Model Context Protocol)

Q35. What is MCP?

Answer

Model Context Protocol is an open standard that allows AI models to communicate with external tools, APIs, databases, and systems through a standardized interface.

Q36. Why MCP?

Answer

Without MCP:

Model ↔ Tool
Model ↔ Tool
Model ↔ Tool

Many custom integrations.

With MCP:

Model
↓
MCP
↓
Tools

One standard protocol.

Q37. MCP Components

MCP Client

MCP Server

Resources

Tools

Prompts

Q38. MCP vs API?

API	MCP
Application communication	AI-tool communication
Custom Integration	Standardized Integration
Not AI-native	AI-native

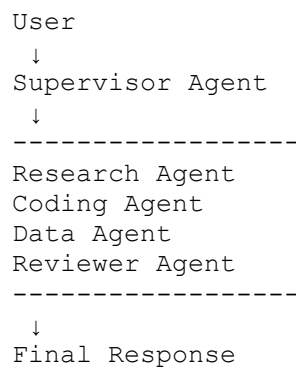
10. Multi-Agent Systems

Q39. What is Multi-Agent AI?

Answer

Multiple specialized agents collaborate to solve complex problems.

Q40. Example Multi-Agent Architecture



Q41. Benefits of Multi-Agent Systems

- Scalability
 - Specialization
 - Parallel Processing
 - Better Accuracy
-

Q42. Challenges

- Coordination
 - State Management
 - Cost
 - Latency
 - Observability
-

11. Azure GenAI

Q43. What is Azure OpenAI?

Answer

Azure-hosted OpenAI models with enterprise security and governance.

Q44. Core Azure AI Services

- Azure OpenAI
 - AI Search
 - AI Foundry
 - Azure Functions
 - Azure Kubernetes Service (AKS)
 - Azure Storage
 - Key Vault
-

Q45. Typical Azure RAG Architecture

Documents

↓

Azure Blob Storage

↓

Azure AI Search
↓
Embeddings
↓
Azure OpenAI
↓
Application

Q46. Why Azure AI Search?

Answer

Provides:

- Vector Search
 - Hybrid Search
 - Semantic Ranking
-

12. API Deployment

Q47. How do you deploy an LLM application?

Answer

User
↓
Frontend
↓
FastAPI
↓
LLM
↓
Vector DB

Q48. Why FastAPI?

- Fast
 - Async
 - Swagger UI
 - Easy deployment
-

Q49. What is an API Gateway?

Answer

Central entry point for APIs.

Examples:

- Azure API Management
 - Kong
 - NGINX
-

13. Docker

Q50. Why Docker for GenAI?

Answer

Ensures:

- Consistency
 - Portability
 - Scalability
-

Q51. Docker Lifecycle

```
Dockerfile
↓
Build
↓
Image
↓
Container
```

Q52. Typical Docker Command

```
docker build -t rag-app .
docker run -p 8000:8000 rag-app
```

Q53. Benefits of Docker in AI

- Environment Consistency
- Dependency Management
- CI/CD
- Cloud Deployment

14. Production GenAI Questions

Q54. What is Observability in GenAI?

Answer

Monitoring:

- Prompts
- Responses
- Latency
- Cost
- Failures

Q55. What is LLM Caching?

Answer

Store previous responses to reduce cost and latency.

Q56. How do you secure enterprise GenAI applications?

Authentication

Authorization

Encryption

PII Masking

Guardrails

Content Filtering

Q57. Common GenAI Architecture Interview Question

Design an enterprise document assistant.

Expected components:

Frontend
↓
FastAPI
↓
Authentication
↓
Azure OpenAI
↓
RAG Layer
↓
Vector DB
↓
Enterprise Documents

Most Important Topics for 2026 Interviews

1. LLM Fundamentals
2. Prompt Engineering
3. Embeddings
4. Vector Databases
5. RAG
6. Agentic AI
7. LangChain
8. LangGraph
9. MCP
10. Tool Calling
11. Multi-Agent Systems
12. Azure OpenAI
13. Azure AI Search
14. FastAPI Deployment
15. Docker
16. AKS
17. Observability (LangSmith, Phoenix)
18. Guardrails
19. Evaluation Frameworks
20. Production GenAI Architecture

LLM Evaluation – Advanced Interview Questions

Q58. Why is LLM Evaluation difficult?

Answer:

Unlike traditional ML:

Traditional ML	LLM
Single correct answer	Multiple valid answers
Accuracy metric	Subjective quality
Structured output	Natural language output

Challenges:

- Multiple acceptable responses
 - Hallucinations
 - Reasoning quality
 - Grounding verification
 - Human judgment required
-

Q59. What are the major categories of LLM Evaluation?

1. Model Evaluation

Evaluate the model itself.

Examples:

- Accuracy
- BLEU
- ROUGE
- Perplexity

2. Application Evaluation

Evaluate the GenAI solution.

Examples:

- Faithfulness
- Relevancy
- Groundedness
- User Satisfaction

3. Agent Evaluation

Evaluate tool usage and reasoning.

Examples:

- Tool Selection Accuracy
- Task Completion Rate
- Plan Quality

Q60. What is Perplexity?

Answer

Measures how well a model predicts the next token.

Lower perplexity = better prediction capability.

Formula:

$$\text{Perplexity} = e^{\text{Loss}}$$

Interview Point:

- Training metric
- Not ideal for evaluating business usefulness

Q61. What is Faithfulness?

Answer

Measures whether the generated answer is supported by retrieved context.

Example:

Retrieved Context:

CEO = John Smith

Generated:

CEO = John Smith

Faithful ✓

Generated:

CEO = Mike Brown

Not Faithful X

Faithfulness is one of the most important RAG evaluation metrics.

Q62. What is Groundedness?

Answer

Checks whether the response originates from provided sources rather than model imagination.

Question:

Who is the CEO?

Context:

CEO = John Smith

Answer:

John Smith

Grounded ✓

Groundedness is commonly used to detect hallucinations in RAG systems.

Q63. What is Answer Relevancy?

Answer

Measures how well the response addresses the user's question.

Question:

What are Azure OpenAI pricing models?

Answer:

Azure OpenAI supports...

Relevant ✓

Answer:

History of Microsoft...

Not Relevant ✕

Q64. What is Context Precision?

Answer

Measures whether retrieved chunks were actually useful.

Formula:

$$\frac{\text{Useful Retrieved Chunks}}{\text{Total Retrieved Chunks}}$$

High precision = minimal noise.

Used heavily in RAGAS evaluation frameworks.

Q65. What is Context Recall?

Answer

Measures whether all required information was retrieved.

Formula:

$$\frac{\text{Relevant Retrieved Chunks}}{\text{Total Relevant Chunks}}$$

Q66. Explain RAGAS.

Answer

RAGAS (RAG Assessment Framework) evaluates:

- Faithfulness
- Context Precision
- Context Recall
- Answer Relevancy

Commonly used for enterprise RAG evaluation.

Q67. What is Hallucination Rate?

Answer

Percentage of answers containing unsupported information.

Formula:

$$\frac{\text{Hallucinated Responses}}{\text{Total Responses}}$$

Lower is better.

Q68. How do you evaluate an Agent?

Metrics

- Task Completion Rate
 - Tool Call Accuracy
 - Plan Quality
 - Number of Iterations
 - Cost
 - Latency
-

Q69. What is LLM-as-a-Judge?

Answer

Using an LLM to evaluate outputs generated by another LLM.

Example:

```
Prompt
Expected Answer
Generated Answer
Judge Model
```

Judge scores:

- Relevance

- Correctness
- Completeness

Q70. What are Offline and Online Evaluation?

Offline

Before deployment.

Examples:

- Benchmark datasets
- Golden datasets
- Human review

Online

After deployment.

Examples:

- User Feedback
- Thumbs Up/Down
- A/B Testing

Q71. What is a Golden Dataset?

Answer

Manually curated questions and expected answers used for evaluation.

Example:

Question
Expected Answer
Retrieved Context

Enterprise GenAI teams typically build hundreds or thousands of golden test cases.

Q72. What metrics should be monitored in production?

Quality

- Faithfulness
- Relevance
- Groundedness

Performance

- Latency
- Throughput
- Token Usage

Business

- User Satisfaction
- Resolution Rate
- Adoption

Transformer Architecture – Advanced Interview Questions

Q73. Why were Transformers introduced?

Answer

RNNs and LSTMs suffered from:

- Sequential processing
- Slow training
- Vanishing gradients
- Poor long-range dependency learning

Transformers replaced recurrence with self-attention and parallel processing.

Q74. What is the Transformer Architecture?

Answer

Original architecture:

Input
↓
Embedding

↓
Positional Encoding
↓
Encoder Stack
↓
Decoder Stack
↓
Linear Layer
↓
Softmax
↓
Output

Introduced in the paper:

Attention Is All You Need.

Q75. What are the key Transformer components?

Embeddings

Positional Encoding

Self-Attention

Multi-Head Attention

Feed Forward Networks

Residual Connections

Layer Normalization

Softmax Layer

Q76. Why is Positional Encoding needed?

Answer

Transformers process tokens in parallel.

Without positional encoding:

I Love AI
AI Love I

would appear identical.

Positional encoding injects sequence order information.

Q77. What is Self-Attention?

Answer

Self-attention allows every token to attend to every other token.

Example:

The animal didn't cross the street because it was tired.

"It" attends to "animal".

This enables long-range dependency learning.

Q78. What are Query, Key and Value vectors?

Answer

Every token generates:

Query (Q)

What am I looking for?

Key (K)

What information do I contain?

Value (V)

Actual information passed forward.

Attention compares Query with Keys and aggregates Values.

Q79. What is Scaled Dot Product Attention?

Answer

Core attention mechanism.

$$Attention(Q, K, V) = Softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Process:

$Q \times K^T$
↓
Scaling
↓
Softmax
↓
Attention Scores
↓
Weighted Sum of V

Q80. Why divide by $\sqrt{d_k}$?

Answer

Prevents very large dot-product values.

Without scaling:

- Softmax saturates
 - Gradients become unstable
-

Q81. What is Multi-Head Attention?

Answer

Multiple attention mechanisms run simultaneously.

Benefits:

- Capture syntax
- Capture semantics
- Capture positional relationships

Each head learns different relationships.

Q82. What is the Feed Forward Network (FFN)?

Answer

After attention:

Attention Output

↓

Linear

↓

ReLU/GELU

↓

Linear

Adds non-linearity and richer representations.

Q83. What are Residual Connections?

Answer

Skip connections.

$\text{Output} = \text{Layer}(x) + x$

Benefits:

- Better gradient flow
 - Faster convergence
 - Deep network stability
-

Q84. What is Layer Normalization?

Answer

Normalizes activations.

Benefits:

- Stable training
 - Faster convergence
 - Prevents exploding activations
-

Q85. Encoder vs Decoder?

Encoder	Decoder
Understands input	Generates output
Bidirectional	Autoregressive
BERT	GPT

Q86. What are Encoder-Only Models?

Examples

- BERT
- RoBERTa
- DistilBERT

Use Cases:

- Classification
 - NER
 - Search
-

Q87. What are Decoder-Only Models?

Examples

- GPT
- Llama
- Claude

Use Cases:

- Text Generation
 - Agents
 - Chatbots
-

Q88. Why are Transformers better than RNNs?

RNN	Transformer
Sequential	Parallel
Slow	Fast
Short Memory	Long Context
Hard to Scale	Highly Scalable

Q89. What is the biggest limitation of Self-Attention?

Answer

Quadratic Complexity.

$$O(n^2)$$

As sequence length grows:

1000 tokens → 1M comparisons
10000 tokens → 100M comparisons

This is why long-context models are expensive.

Q90. What modern innovations improve Transformers?

Flash Attention

RoPE (Rotary Positional Encoding)

KV Caching

Mixture of Experts (MoE)

Linear Attention

Sparse Attention

Used by modern LLMs to improve speed and scalability.

High-Frequency Interview Questions

1. Explain Self-Attention mathematically.
2. Why Q, K, V?
3. Why Multi-Head Attention?
4. Why Positional Encoding?
5. Transformer vs LSTM.
6. Encoder-only vs Decoder-only.
7. How does GPT work internally?
8. How does BERT differ from GPT?
9. Explain RAGAS metrics.
10. Faithfulness vs Groundedness.
11. LLM-as-a-Judge.

12. How do you evaluate an agentic workflow?
13. How would you build a GenAI evaluation framework for production?
14. How do you detect hallucinations?
15. How do you evaluate a multi-agent system?